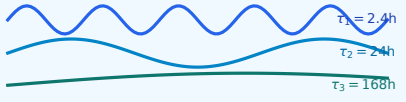
C-STGB v2: End-to-End Architectural Dataflow & Neuro-Symbolic Tensor Pipeline

Six Interconnected Modules: Continuous Time -> Edge Denoising -> Latent SMOTE -> Evidence Fusion -> Bayes Head -> Conformal CRC

TIER 1: SPATIO-TEMPORAL NEURAL ENCODING & ADVERSARIAL DENOISING

Module 1: Continuous Temporal Encoder

Tri-Band Harmonic Positional Encoding:



Self-Exciting Hawkes Rate $\lambda_u(t)$:



Output Tensor: $\mathbf{h}_u^{(0)} \in \mathbb{R}^d$ **(Temporal Seed)**

Module 2: Edge-Trust Filter & GNN

Learnable Edge-Trust Gating Mechanism:



Spatio-Temporal Attention Aggregation:

$$\tilde{\mathbf{h}}_u = \sum_{v \in \mathcal{N}_u} \alpha_{vu} \mathbf{W}_v \mathbf{h}_v^{(l-1)} \hat{g}_{vu}$$

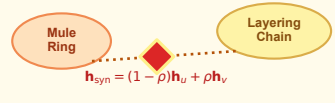
Dirichlet Energy Preserved: $\mathcal{E}(\mathbf{H}) > 0.75$

Threshold Floor: $\delta_{\text{floor}} = 0.10$

Structural Embedding: $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$

Module 3: Latent GraphSMOTE

Latent Typology Manifold Synthesis:



Bilinear Link Gen & Physical Bounds:

$$\hat{A}_{\text{syn},w} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_e \mathbf{h}_w)$$

Physical Invariant Guardrail: $\Phi_{\text{flow}} \approx 1.0$

Offline Training Quarantined (0% Test Leak)

Balanced Training Graph: $\tilde{\mathcal{G}}_{\text{train}}$ **(+9.92 pp Lift)**

Offline Augmentation

Structural Embedding $\mathbf{h}_u^{(L)} \in \mathbb{R}^d$ | **12-D Canonical Invariants** $\mathbf{z}_{\text{inv}} \in \mathbb{R}^{12}$ **(Hydrodynamic Mass-Flow + Hawkes + Taint)** | **Edge Gates** $\hat{g}_{vu} \geq 0.10$

TIER 2: EVIDENCE FUSION, BAYESIAN OPTIMIZATION & CONFORMAL RISK CONTROL

Module 4: Evidence-Adaptive Fusion

Adaptive Inductive Gating Unit $\alpha_u \in [0, 1]$:

$$\mathbf{h}_u^* = \alpha_u \mathbf{h}_u^{(L)} + (1 - \alpha_u)\mathbf{z}_{\text{inv}}$$

where $\alpha_u = \sigma(\mathbf{w}_\alpha^T[\mathbf{h}_u^{(L)} \| \mathbf{z}_{\text{inv}}])$

Cold-Start (deg <= 1)
$\alpha_u \to 0$ (Inv Dominates)
89.60% Cold-Start F1

Dense Graph (deg >> 1)
$\alpha_u \to 1$ (GNN Dominates)
Multi-Hop Topology

12-D Canonical Invariant Inputs:

Φ_{flow} **(Kirchhoff Mass-Flow) +** λ_u **(Hawkes)**

$\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}$ (Causal Taints) + Moments $(\mu, \sigma^2, \gamma, \kappa)$

Zero Retraining across Heterogeneous Ledgers

Fused Evidence Vector: $\mathbf{h}_u^* \in \mathbb{R}^d$

Module 5: Cost-Sensitive Bayes Head

Asymmetric Focal Tversky Loss (Eq. 3.28):

$$\mathcal{L}_{\text{task}} = 1 - \frac{\sum p_i y_i + \varepsilon}{\sum p_i y_i + 0.85\sum(1-p_i)y_i + 0.15\sum p_i(1-y_i) + \varepsilon}$$

5.67x FN Gradient Magnification ($\beta_{\text{FN}} = 0.85$ vs $\beta_{\text{FP}} = 0.15$)



Multi-Task Representation Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{asym}} + \lambda_1 \mathcal{L}_{\text{edge}} + \lambda_2 \mathcal{L}_{\text{flow}}$$

Non-Conformity Scoring: $s_i = 1 - \hat{p}_i(y)$

Calibrated Posterior for Distribution-Free CRC

Calibrated Risk Score: $\hat{p}_u \in [0, 1]$

Module 6: Conformal CRC Arbiter

Class-Conditional Quantiles $(1 - \alpha \geq 99\%)$:

$$\hat{q}^{(0)} = \text{Quantile}\left(s_i^{(0)}, \frac{\lceil (n_0+1)(1-\alpha_0) \rceil}{n_0}\right)$$

$$\hat{q}^{(1)} = \text{Quantile}\left(s_i^{(1)}, \frac{\lceil (n_1+1)(1-\alpha_1) \rceil}{n_1}\right)$$

Tier 1: Quarantine Hold **0.18% (18/10k)**
$\Gamma(u) = \{1\}$ | Auto Wire Freeze & FinCEN Form 111

Tier 2: Compliance Desk **0.49% (49/10k)**
$\Gamma(u) = \{0, 1\}$ | **-99.51% Human Review Queue Cut**

Tier 3: Straight-Through (STP) **99.33% (9.9k/10k)**
$\Gamma(u) = \{0\}$ | **0.45 ms Real-Time Clear (<10ms SLA)**

Deterministic Decision Set: $\Gamma(u) \subseteq \{0, 1\}$